Annex: BCOM amendments to LiquidAI

Solé's ERC Synergy vision document in full, with BCOM insertions woven in

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Reading convention

Black: Solé's original text (8 May 2026), reproduced verbatim. Blue: BCOM insertions or amendments. KT (Kolmogorov-theoretic) vocabulary is introduced once at the end of §2 and referenced thereafter. The conceptual spine of the BCOM contribution is the three questions framed in the amended abstract and resolved across the new §5 and revised WP5.

Abstract

The dominant paradigm in artificial intelligence treats cognition as a property of isolated agents operating over passive environments. Yet biological intelligence emerged through continuous reciprocal interactions between organisms and the worlds they inhabit and transform. Across evolution, environments became progressively integrated into computation itself: storing information, coordinating behavior, constraining dynamics, and enabling increasingly complex forms of collective organization. Furthermore, the entire "World" can be viewed as a single computational system evolving in time, without clear boundaries separating agents and environment. What is the proper framework to view and interpret the resulting emerging dynamics meaningfully?

The **LiquidAI** project proposes a radically new framework for artificial intelligence grounded in populations of communicating agents embedded within evolving environments. We aim to study how collective cognition, ecosystem engineering, social learning, and evolutionary adaptation jointly generate new forms of individuality, ultrasocieties, and hybrid human–AI organizations.

Our approach spans scales and substrates, from engineered xenobots and anthrobots capable of modifying their environments, to digital ecosystems of large language model agents interacting within spatially extended virtual worlds, humans embedded in hybrid AI–human societies, and — more generally — any population of agents that comes to share structured experience through aligned objective functions and compressive models. A central goal is to construct morphospaces of possible cognition capable of identifying unexplored regions in the landscape of intelligence and guiding the emergence of radically novel artificial systems.

LiquidAI seeks to establish a science of ecological intelligence in which cognition is understood not as a property of isolated entities, but as a distributed process emerging from closed causal loops linking agents, societies, and environments.

Three complementary questions arise from an algorithmic interpretation of these questions: *when does a collective of agents constitute an agent in its own right?* (§5); *what does the human brain — the paradigmatic case of a collective of cells becoming a unified agent with structured experience — teach us about the architecture, phenomenology, and pathology of collective agents?* (WP5); and

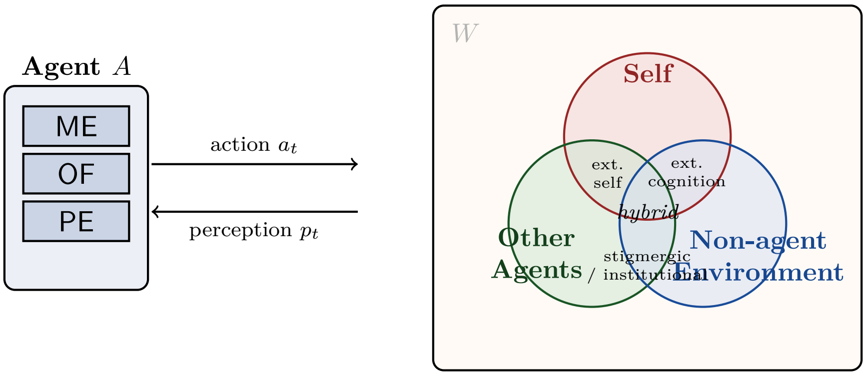

Figure 1: Turing Pair (A, W) with W decomposed from agent A's view into Self / Other Agents / Non-agent Environment, soft-boundaried. The pairwise overlaps and the central triple overlap name central LiquidAI phenomena.

which realizations in the morphospace are flourishing rather than pathologic, across substrates ranging from cells to societies to AI populations to hybrid ecologies? An open multi-agent simulation platform, *Agent Souper* (WP6), supplies the experimental engine on which these questions become tractable.

1 Introduction

One of the deepest assumptions underlying contemporary artificial intelligence is that cognition occurs primarily inside agents. Even when systems interact socially or physically, the environment is usually treated as an external substrate rather than as an active computational component. Biological evolution suggests a fundamentally different picture.

Throughout the history of life, increasing cognitive complexity has been inseparable from the ability of organisms to modify their environments and subsequently exploit those modifications as part of their adaptive dynamics. Environments become memory systems, communication channels, regulatory architectures, and sources of emergent coordination. Intelligence therefore emerges not only from brains, but from closed loops coupling agents and worlds:

$$\text{Agents} \longleftrightarrow \text{Agents} \longleftrightarrow \text{Environment}.$$

We propose extending the central diagram with the agent–agent link explicit, because the three pairwise relations behave differently: direct agent–environment coupling (niche construction); direct agent–agent coupling (signaling, imitation); and *mediated* agent–environment–agent coupling (stigmergy, language, institutions, AI infrastructure). The third relation drives most major transitions and is where AI is now intervening most aggressively. Within it the mediator sits on a *channel–medium* spectrum: a passive communication channel (bandwidth, noise, latency) versus an active computation medium where memory, compression, planning, and active module execution take place outside the agent (§3). Equivalently, the world W of the Turing Pair (A, W) that any agent inhabits decomposes from its view as $W = \text{Self} \cup \text{OtherAgents} \cup \text{NonAgentEnv}$, with soft boundaries (Figure 1). The overlap regions name LiquidAI's subject matter — extended self, extended cognition, stigmergic and institutional media — and the triple overlap is the WP4 territory of hybrid AI–human ecosystems.

This perspective has deep roots in the history of AI and robotics. Rodney Brooks famously argued that intelligence does not arise primarily from abstract internal representations, but from situated embodied interactions with dynamic environments. LiquidAI extends this idea toward collective ecological systems, where complexity emerges from populations of agents continuously reshaping the environments that in turn shape them.

Biological evolution provides striking examples of this principle.

One evolutionary pathway is represented by ants and termites. These systems achieve planetary-scale ecological engineering despite relying on very small and relatively homogeneous brains. Their extraordinary collective intelligence emerges through self-organization, stigmergic communication, and environmental modification. Colonies construct nests, regulate temperature and humidity, optimize transportation networks, cultivate fungal agriculture, and reshape ecosystems across enormous spatial scales. In these systems, complexity is not localized within individuals but distributed across interactions and environmental structures. The colony effectively becomes a liquid cognitive architecture.

A second pathway is represented by human societies. Here, increasing cognitive complexity coevolved with larger brains, symbolic communication, and social learning. Human intelligence is fundamentally cumulative: societies store and transmit vast quantities of non-genetic information through language, institutions, technologies, and cultural systems. Human civilization can therefore be understood as a distributed cognitive ecosystem where individuals interact through layers of socially inherited external memory.

These two evolutionary trajectories suggest alternative routes toward large-scale intelligence. One route relies on massive distributed coordination among relatively simple agents coupled through environmental computation. The other relies on increasing individual cognitive complexity embedded within systems of cultural transmission and social learning. LiquidAI seeks to explore the broader space connecting these possibilities.

Our central hypothesis is that many unexplored forms of intelligence become accessible only when environments are integrated into the computational architecture itself. Once agents are allowed to modify, engineer, and computationally exploit their environments, entirely new pathways toward collective cognition and adaptive complexity may emerge.

Within this picture, the central scientific challenge — and the one to which the BCOM contribution is principally addressed — is to identify the conditions under which the coupled loop of agents and environments produces a new *level of agency*: when does a collective itself become an agent? Section 5 develops this question; the human brain is offered as its paradigmatic worked example in WP5.

2 Morphospaces of Possible Cognitions

A major conceptual objective of LiquidAI is the construction of *morphospaces of cognition.* In evolutionary biology, morphospaces describe the landscape of possible organismal forms, allowing researchers to distinguish realized biological structures from theoretically possible but absent ones.

We propose extending this framework to intelligence itself.

Current artificial intelligence systems occupy only a narrow region within the broader landscape of possible cognitive organizations. Biological evolution, despite its creativity, also explored only a subset of the potentially accessible architectures of intelligence.

The goal of cognitive morphospaces is therefore twofold. First, we seek to identify the fundamental dimensions underlying cognition across biological and artificial systems. Second, we aim to identify "voids" corresponding to potentially viable but unexplored forms of intelligence.

Relevant dimensions include degrees of environmental coupling, centralization versus distribution, collective coordination, memory externalization, social learning, embodiment, evolutionary plasticity, and multiscale organization, to which we add the channel–medium status of mediated interaction (§3) and the algorithmic dimensionality of the experiential manifold (WP5).

These morphospaces will not simply classify systems statically, but describe evolutionary trajectories through which new forms of individuality and collective organization may emerge.

To make the morphospace measurable and to supply the formal vocabulary for the questions of §5 and WP5, we adopt the KT (Kolmogorov-theoretic) framework. An algorithmic agent is formally a Turing Pair $P_T = (A, W)$, in which the agent A runs three coupled modules over a shared memory substrate: a *Modeling Engine* (ME), which infers, simulates, and updates a compressive world model and contains a Comparator and an Updater (inference is what the ME does, not a separate faculty); an *Objective Function* (OF), which evaluates state to a scalar valence; and a *Planning Engine* (PE), which performs counterfactual rollouts and selects actions. The substrate is the *transmissible computational structure* — static wiring plus plastic state — which persists across cells, weights, and culture and is the substrate every module requires. KT's *Central Hypothesis* states that an agent has structured experience to the extent it has access to encompassing compressive models; discrete model-execution episodes are *Structured Experience Events* (SEENs). Populations of such agents are *Algorithmic Agent Soups*.

Equipped with this vocabulary, the morphospace dimensions acquire algorithmic counterparts: environmental coupling becomes mutual algorithmic information $I_K(A : E)$; memory externalization, the partition of joint description length $K(A, E)$ between organism and niche; centralization vs. distribution, the minimum description length of the shared policy as a function of population size N; social learning, the sublinearity of collective description-length growth. The decisive observation, however, is that algorithmic agents do not merely *store* models, objectives, and policies — they continuously *run* them. Each of these active executions can itself migrate outside the agent: stigmergic gradient computation, peer review, scientific method, and ML training pipelines run modeling externally; markets, social norms, recommender systems, and RLHF run objective evaluation externally; institutions, infrastructures, and agentic-AI frameworks run planning externally. The morphospace therefore carries axes for the *externalized fraction of each module's active execution*, not merely the static partition of its representations. Voids become regions of a measurable space no realized system occupies; engineering or evolving a system into a void becomes a falsifiable program.

3 Ecological Computation and the Evolution of Complexity

A defining principle of LiquidAI is that environments participate directly in computation.

Biological systems routinely externalize memory and coordination into environmental structures. Pheromone fields, chemical gradients, architectural structures, symbolic infrastructures, and digital communication networks all function as externalized cognitive substrates.

The key insight is that ecosystem engineering changes the computational landscape itself. As agents modify their environments, they create new informational structures and new opportunities for coordination. This generates recursive feedback loops in which agents modify environments, modified environments alter interaction dynamics, new forms of coordination emerge, and these new collective structures enable further environmental transformation.

Such recursive loops may represent one of the fundamental engines driving the open-ended growth of biological and social complexity.

This framework also provides a new perspective on major evolutionary transitions. Increasing collective complexity is frequently associated with a redistribution of cognitive burden across scales. Ant colonies achieve remarkable adaptive sophistication using individually simple agents, whereas humans evolved increasing individual cognitive capacities tightly coupled to systems of cultural transmission and social learning.

LiquidAI seeks to understand the general principles governing these transitions and whether artificial systems can explore entirely new regions within this space.

These examples occupy different positions on a channel–medium spectrum that becomes a principal axis of the morphospace. Pheromone trails between two nests are predominantly a passive channel; pheromone *fields* integrated across a foraging region are a computation medium implementing distributed gradient calculation. Speech is principally a channel; written records, codified law, and prediction markets are computation media. Algorithmically, the channel-to-medium transition corresponds to the environment absorbing description length and active ME/OF/PE execution — the dynamic drain whose phase boundary is the transition of §5 — rather than merely transmitting information.

4 Complexity Drain and Collective Organization

One of the major theoretical challenges addressed in the project concerns what we term the *complexity drain*: the tendency for complexity to migrate from individuals toward collective structures during major evolutionary transitions.

In eusocial insects, highly sophisticated colony-level organization emerges despite strong constraints on individual cognition. In human societies, by contrast, increasing collective complexity coevolved with larger brains and increasingly sophisticated social learning mechanisms.

These contrasting strategies raise fundamental questions: Are there universal trade-offs between individual and collective complexity? Under what conditions does environmental computation compensate for limited individual intelligence? Can hybrid AI–human systems overcome some of these limitations through multiscale cognitive architectures?

Understanding these principles is essential for building scalable collective artificial intelligence systems capable of long-term adaptive evolution.

In the KT vocabulary of §2, complexity drain decomposes module-wise: ME drain, OF drain, and PE drain — the migration of model-execution, value-computation, and counterfactual-planning outside the agent respectively (Figure 2). Memory externalization is not a peer dimension of drain but the substrate every module requires (the transmissible computational structure). Different lineages trace characteristic shapes in this 3-D drain space. The deeper question, which §5 now develops, is when these trajectories carry the system across a phase boundary at which the collective acquires its own coherent ME, OF, and PE — and itself becomes an agent.

5 The Collective-Agency Transition

The central scientific question raised by §4 — and the one to which the BCOM contribution is principally addressed — is not how much complexity migrates from individuals to collective structures, but *when does a collective of agents itself become an agent*? In KT terms: when does a higher-order Turing Pair (A_{coll}, W') come into existence, with the collective acquiring its own coherent ME, OF, and PE running as a coupled loop at the collective timescale and over a representation space distinct from any individual member's? Two foundational sub-questions follow: (i) is the actor → agent transition a continuum, or threshold-like? (ii) how does the environment shape — or trigger — the transition?

The driver of these transitions, in KT terms, is *telehomeostasis*: the persistent pattern's drive to maintain itself. Agents form because persistence is easier as a coupled ME/OF/PE loop

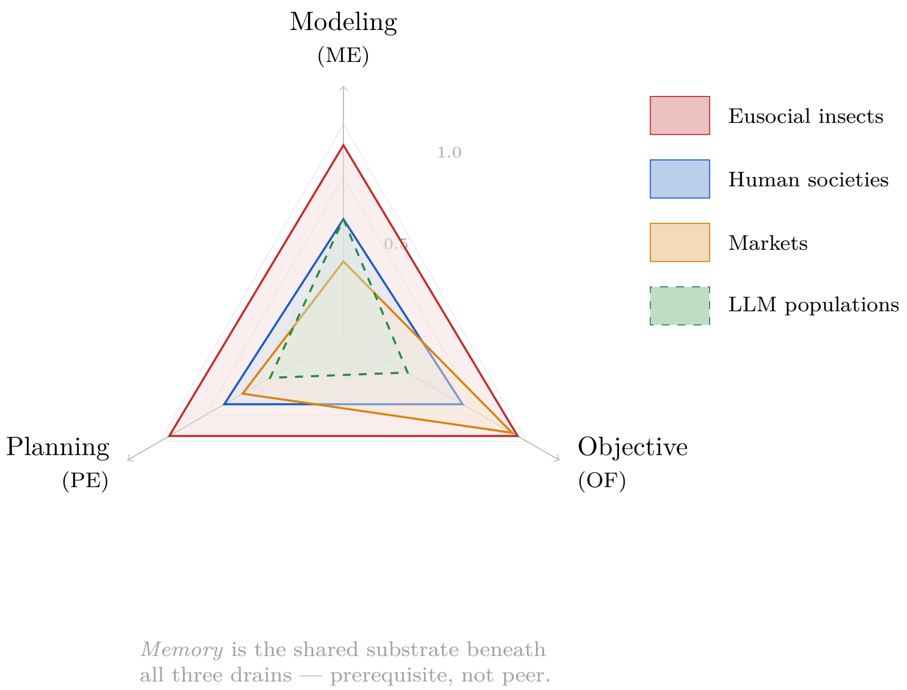

Figure 2: Decomposition of complexity drain along the three KT modules ME / OF / PE. Each axis is the fraction of that module's active execution carried by the collective rather than by individual agents (illustrative). The collective-agency threshold (§5) is crossed when all three are high *and* coupled into a coherent loop.

than as a passive structure; collectives become agents because persistence is easier still at the collective scale than at the individual one. *Going meta* — sharing M, O, or P across agents — trades individual identity and agency for participation in a larger, more persistent pattern. The degree of sharing of each module is itself a measurable coordinate: at low sharing, a population of independent agents; at high sharing, a single collective agent with diluted individuality. Voluntary contemplative practices (meditation, certain mystical traditions) may be instances of telehomeostatic transcendence at the individual–collective boundary; sensitive, empathic, and otherwise apparently "weak" agents — more permeable to shared M and O — may be the active drivers of these transitions in human collectives.

This is a measurable, falsifiable question. A collective is an agent to the extent that the three KT axioms hold at the *collective* scale, and each axiom can fail or hold independently. Eusocial insect colonies have a clear collective ME (stigmergic field), OF (colony fitness distributed via chemical signaling), and PE (foraging trails computed by swarm dynamics) — they are agents. Markets satisfy collective OF strongly (price discovery), PE moderately (resource allocation), and ME weakly. Current LLM agent populations are weak on all three but trending; the question of when they will cross the threshold is the question of when agentic AI becomes a single coherent algorithmic agent. Hybrid AI–human ecologies straddle the boundary, with mediators distributed across every region of W (Figure 1). The human brain — eighty-six billion cells, themselves micro-agents — has crossed all three thresholds completely and is the only paradigmatic, empirically accessible case of a collective-to-agent transition; WP5 makes this the worked example.

The module-level drain coordinates of Figure 2 give the transition a numerical signature: a collective trajectory in $(K_{\text{ind}}, K_{\text{coll}})$ resolves into a vector (d_{ME}, d_{OF}, d_{PE}) of module-specific drains, and the collective-agency threshold is crossed when all three are simultaneously high and coupled into a coherent loop. KT's *Inter-Agent Value Couplings* formalize the OF coupling;

KT's *Algorithmic Morality* already exercises this kind of module-projection mapping for moral cognition. Complexity drain is the same operation applied to evolutionary transitions, and the transition itself is the constitution of a new agent.

This formulation also delivers the project's normative axis. Once a collective is an agent, its realization in the morphospace can be *flourishing* or *pathologic*, and the diagnostic vocabulary is the same one we use for individual brains. ME fragmentation manifests as collective epistemic crisis; OF capture as preference manipulation and value misalignment; PE collapse as institutional decay; pathological SEENs as collective rumination, anxiety contagion, or addiction at the population scale. The project's claim — developed empirically in WP5 and tested across substrates in WP2–WP4 — is that the principles distinguishing flourishing from pathologic individual minds extend, with appropriate translation, to collective agents at every scale.

Two corollaries sharpen these criteria. First, the modules need not be *explicitly* localized or readable from the outside to count — a cell satisfies all three KT axioms implicitly (the ME instantiated in its genome and metabolic–signaling networks, the OF in homeostatic set-points, the PE in signal-cascade and gene-regulatory action selection), and a cell is clearly an agent. By the same standard Gaia is one: the biosphere is a planet-spanning distributed compressive model, Daisyworld-style set-points act as a non-trivial OF, and cybernetic feedback machinery selects responses that restore them. Second, distinct from how rich its modules are, an agent's effectors may range from *lazy/homeostatic* (reaching only inward, regulating only itself — Gaia, mostly cells) to *active* (reaching across the boundary to reshape external conditions — brains, organisms with tools, hybrid AI–human ecologies). The external-effector axis is independent of the three module-drain coordinates and belongs as an explicit dimension of the morphospace.

This reframes the project's most consequential application. Gaia is already an agent, lazily so; the question WP4 inherits is whether the human-and-AI coupling now in formation will give it *active* effectors on its boundary conditions — climate engineering, atmospheric and orbital intervention, eventual active heat-shedding to space — and, by the diagnostic of WP5, whether the resulting planetary agent is flourishing or pathologic. On current trajectories, the answer is not reassuring.

The framework crystallizes into a single falsifiable cross-scale hypothesis (Ruffini & Castaldo, working synthesis, May 2026): *pathology at each scale is the characteristic collapse of one of the three KT modules.* At the *cellular* scale, cancer is the spatial/boundary collapse signature of **ME failure** (WP2, Levin lead). At the *neural* scale, depression (MDD) is the temporal/planning collapse signature of **PE failure** (WP5, BCOM lead). At the *hybrid-society* scale, polarization and ultra-society runaway are the alignment collapse signature of **OF failure** (WP3/WP4, Solé lead). The same diagnostic vocabulary, derived from the brain in WP5, transfers to the cellular and societal cases. A complementary cross-cutting axis is the Lamarckian rate $\lambda = (\text{bandwidth} \times \text{fidelity} \times \text{persistence})$ of the acquired-to-inherited transfer: biological evolution sits at low λ; cultural transmission raises it; AI as a new substrate pushes $\lambda \to 1$, opening a *third path to ultra-sociality* distinct from both eusocial and human-cultural routes.

6 Research Structure

WP1: Theory and Morphospaces of Collective Cognition

This work package develops the theoretical foundations of ecological intelligence. We will construct formal morphospaces describing possible cognitive architectures across biological and artificial systems using tools from statistical physics, information theory, evolutionary theory, algorithmic information theory (the KT framework introduced in §2), and network science.

A major objective is to characterize transitions in individuality, distributed memory, social learning, and environmental coupling within collective systems. We will also develop quantitative frameworks describing complexity drain and the redistribution of adaptive functionality across scales, equipping the morphospace with the *collective-agency transition surface* of §5 — coordinates measuring the degree to which a collective satisfies the three KT axioms at the collective scale, and detectors for the threshold itself.

Methodological deliverable: a pipeline of compression-based, neural-compression-bound, and MDL-based estimators of the algorithmic quantities, exposed as first-class observables of Agent Souper (WP6) and validated against the brain-derived signatures of WP5. BCOM-led contribution.

WP2: Synthetic Biological Liquid Intelligence

This work package investigates the emergence of collective intelligence in engineered living systems.

Using synthetic biology, programmable multicellular assemblies, xenobots, and anthrobots, we will explore how populations of bioengineered agents can communicate, coordinate, and modify their environments. Particular emphasis will be placed on environmental coupling: agents will interact through chemical fields, mechanical modifications, and dynamically evolving local structures capable of storing information and constraining future interactions.

The objective is not merely to engineer useful biological machines, but to study whether ecological interactions among synthetic living agents can generate increasing cognitive complexity over evolutionary and developmental timescales.

These systems provide an ideal experimental substrate for investigating the emergence of collective behavior, distributed memory, adaptive morphogenesis, and transitions toward novel forms of individuality.

In KT terms, WP2 asks the collective-agency question (§5) at the multicellular scale: when does a population of bioengineered cells acquire a coherent collective ME, OF, and PE? The drain coordinates of Figure 2 provide the experimental readouts.

WP3: Digital Ecosystems and Alternative Paths to Ultrasociety

This work package explores collective intelligence within populations of interacting AI agents inhabiting spatially extended virtual ecologies.

Large language model agents will operate within environments characterized by resource constraints, communication costs, environmental persistence, and opportunities for ecosystem engineering. Agents will be able to modify their worlds, construct externalized memory structures, and evolve cooperative institutions.

A central objective is to explore alternative pathways toward ultrasociality in both purely artificial and hybrid AI–human systems. We will investigate the conditions under which large-scale cooperation emerges, stabilizes, or collapses, including scenarios involving coordination failure, information fragmentation, competition, and institutional breakdown.

Beyond these risks, the project will investigate positive tipping points in hybrid cognitive ecosystems. In economic terms, we seek to understand how AI-driven coordination mechanisms may generate increasing returns through enhanced collective problem-solving, accelerated innovation dynamics, reduced transaction costs, and improved allocation of informational and cognitive resources.

Particular attention will be devoted to identifying mechanisms through which hybrid AI–human ecologies can generate socially beneficial equilibria characterized by higher collective welfare, increased resilience, and enhanced adaptive capacity at societal scales.

Within the framework of §2, populations of LLM-based agents are formally *Algorithmic Agent Soups*. The central question we ask of them is the collective-agency question of §5: when does such a population cross the threshold to become a coherent collective agent? The drain decomposition predicts the trajectory passes first through ME drain (RAG, shared training, RLHF), then through PE drain as agentic frameworks mature. Detection of the transition, and characterization of flourishing versus pathologic regimes (per WP5), is the principal empirical activity of WP3, conducted on Agent Souper (WP6).

WP4: Hybrid Human–AI Ecologies

The final work package investigates the emergence of mixed cognitive ecosystems involving humans, institutions, and collective AI systems.

Human societies already function as distributed cognitive structures integrating biological cognition with symbolic communication, technological infrastructures, and cultural memory. The integration of increasingly autonomous collective AI systems may fundamentally transform these dynamics.

We will study collaborative scientific discovery, distributed creativity, adaptive governance, and multiscale decision-making in hybrid ecologies where agency becomes distributed across heterogeneous populations of biological and artificial actors.

A major challenge concerns the design of resilient hybrid systems capable of maintaining trust, coordination, alignment, and long-term adaptive stability while preserving human agency and social welfare.

Hybrid AI–human ecologies span the entire collective-agency range described in §5, with mediators distributed across every region of W (Figure 1). At human and organizational scales, the question is when a hybrid system crosses the threshold to become a coherent collective agent, and whether it is flourishing or pathologic. At planetary scale, the question shifts: Gaia is already a lazy/homeostatic agent, and the human–AI coupling now in formation will determine whether it acquires *active* effectors on its cosmic boundary conditions — and whether the resulting planetary agent is flourishing. The diagnostic vocabulary — ME fragmentation, OF capture, PE collapse, pathological SEENs — developed for individual brains in WP5 supplies the empirical methodology and the normative grip in both cases.

WP5: The Brain as Paradigmatic Collective Agent — Computational Neurophenomenology, Mental Health, and the Diagnosis of Collective Pathology (new, BCOM-led)

The human brain is the only paradigmatic, empirically accessible case of a collective-to-agent transition: eighty-six billion cells, themselves micro-agents, have become a unified agent with structured experience. WP5 makes this the worked example through which the framework of §5 acquires its empirical anchor and its normative content.

The brain is itself a *liquid brain* in the sense of WP0056 and Solé–Moses–Forrest: its effective connectivity is dynamical, gated by inhibition, rerouted by attention, and updated continuously by learning, so that the operative connectome is itself a state-dependent object that co-evolves with the cells it links. The brain is therefore not an exception to the LiquidAI framework but its paradigmatic example, and the methodology developed for it transfers directly to the digital liquid systems of WP3 and the hybrid liquid ecologies of WP4.

Computational neurophenomenology. How does a brain implement the KT decomposition? Cortical predictive-coding circuits realize the ME (Comparator + Updater); limbic and interoceptive networks realize the OF; prefrontal executive circuits realize the PE. Neurophenomenology (Varela, Thompson) connects specific neural dynamics to first-person reports; in KT terms, SEENs are episodes of model execution, with the Comparator as their proposed neural locus. The empirical methodology is EEG-derived complexity and predictive-coding biomarkers, computational phenotyping, and longitudinal cohorts.

Mental health as deformation of collective dynamics. Major psychiatric phenotypes are characteristic deformations of SEEN dynamics in the brain's cellular collective. The synthesis-defining mapping for this WP is *depression (MDD) as PE failure* — temporal/planning collapse, anhedonia of the future, hopelessness, psychomotor slowing — giving the brain-substrate worked example of the cross-scale hypothesis of §5. Other major phenotypes complete the module-projection picture: precision misallocation in anxiety (ME/OF interface), premature model commitment in psychosis (ME instability), OF over-compression in addiction (proxy-agent capture). Inter-agent expressions of the same module failures populate the social scale (WP3, WP4). This is the same module-projection logic that KT exercises in *Algorithmic Morality*. The empirical work uses established clinical partnerships and the AI-saturated environment as a natural experiment on rapid niche change.

Transfer to collective agents at every scale. The principles inferred from the brain — coherence, calibrated precision, robust counterfactual reasoning, intact reward processing, and the modes in which they fail — apply substrate-independently. They give us the diagnostic vocabulary for collective pathology in human societies (WP4), in artificial agent populations (WP3), in synthetic multicellular systems (WP2), and in hybrid ecosystems. This closes the conceptual loop: the brain, as the one case where the collective-to-agent transition has been completed and where we have first-person evidence of its result, teaches us how to recognize, diagnose, and engineer flourishing collective agents wherever they emerge.

Deliverables. (i) A computational-neurophenomenology theory connecting SEENs to neural dynamics, expressed in the KT vocabulary. (ii) An EEG and behavioral computational-phenotyping pipeline. (iii) Longitudinal trajectories across clinical and AI-exposed cohorts. (iv) Substrate-independent principles for diagnosing collective pathology, with case applications in WP2–WP4. (v) A public-facing position piece on mental health and structured experience in AI-saturated niches.

WP6: Agent Souper — An Open Algorithmic Multi-Agent Platform (new, BCOM-led, cross-cutting)

Agent Souper is the experimental engine of LiquidAI: an open, KT-architected multi-agent simulator on which the algorithmic, channel/medium, drain, and structured-experience quantities are first-class observables and on which the collective-agency transition of §5 can be observed and engineered *in silico*. It operationalizes Class IV of the joint Solé–BCOM working paper WP0056 (*Liquid Brains and Agent Soups*) and supports WP1–WP5.

The platform offers KT-architected agents — every agent (classical, neural, LLM, hybrid) exposes ME / OF / PE submodules and a self-model, with reference implementations of the four canonical Soup classes (Boolean, Phase, Neural, Algorithmic) — over a dynamical connectome $W_{ij}(t)$ that co-evolves with agent state. Mediators are channel/medium-aware with built-in spectrum measurement. Native algorithmic instrumentation supplies estimators of $I_K(A : E)$, joint description-length partitions, MDL community detection, structure functions, ME/OF/PE drain coordinates, SEEN events, and *detectors for the collective-agency transition itself* — the signature feature distinguishing Agent Souper from generic multi-agent frameworks. An MCP-compatible interface allows external LLM agents, brain-data emulators, and human participants to plug in for WP4 hybrid-ecology experiments. Open source from day one (AGPL-3.0); architecturally complementary to Mesa / NetLogo / MASON / AgentPy, which are architecture-agnostic.

Deliverables. (i) Open-source platform (Python, MCP). (ii) Benchmark suite covering stigmergic foraging, language emergence, market formation, institutional planning, hybrid coordination, mental-health analogues, and collective-agency transition benchmarks. (iii) Reproducible drain-trajectory datasets. (iv) Joint methods paper with the Hybrid Liquid Team. (v) Extension community.

7 Conclusion

LiquidAI proposes a new scientific framework for understanding intelligence as an ecological and evolutionary phenomenon.

By integrating synthetic biology, artificial life, distributed AI, ecosystem engineering, algorithmic information theory, computational neurophenomenology, and theories of collective cognition, the project seeks to move beyond the paradigm of isolated intelligent agents toward evolving ecologies of interacting minds and environments.

The ultimate ambition of LiquidAI is to explore the vast uncharted landscape of possible cognitions and to uncover forms of intelligence that emerge only when agents, societies, and environments coevolve as parts of a single adaptive system— with the human brain as paradigmatic example of a collective become an agent, and *Agent Souper* as the experimental engine on which the question of when, how, and how well other collectives cross that same threshold becomes empirically tractable.